Research Report

Executive summary

Objective: Analyze customer churn (Exited) in Churn_Modelling.csv, identify why customers leave, prioritize high-risk segments, and provide actionable recommendations + implementation roadmap.

Dataset: 10,000 customers. Observed churn rate: 20.37% (2,037 customers exited).

Key model & metric highlights - Model: Random Forest classifier trained on demographic and account features. - AUC: 0.851 (good ranking ability). - Test accuracy: 85.95%. - Churn (Exited=1) class: precision 0.76, recall 0.45, F1 0.57 — model is conservative (many churners missed). - Most important features (by feature importance): Age (24.0%), EstimatedSalary (14.7%), CreditScore (14.4%), Balance (14.1%), NumOfProducts (13.0%), Tenure (8.1%), IsActiveMember (3.9%), Geography (3.9%).

Primary findings (evidence-based) 1) Higher churn among older customers - Mean age: churners = 44.84 yrs vs non-churners = 37.41 yrs. Age is the top predictor. - Interpretation: older customers may be less engaged digitally, more sensitive to service issues, or shifting to competitors.

2) Higher balance is associated with churn - Mean balance: churners = $91,108 vs non-churners = $72,745. - Interpretation: customers keeping more funds are more valuable but also more likely to leave — critical segment to retain.

3) Lower activity strongly correlated with churn - IsActiveMember mean: churners = 0.361 vs non-churners = 0.555. Inactive customers are much more likely to leave. - Suggested cause: lack of product/feature engagement or poor onboarding/relationship management.

4) Geography: Germany concentrated churn - Churn rates by country: Germany ≈ 32.5%, France ≈ 16.2%, Spain ≈ 16.7%. - Germany requires urgent investigation and intervention.

5) Gender differences - Female churn rate ≈ 25.1% vs Male ≈ 16.5%. - Further segmentation needed to understand whether product fit, communication, or experience differs by gender.

6) Other signals - Churners slightly lower average credit score (645 vs 651) and slightly fewer products on average (1.48 vs 1.54). - Tenure not a strong separator in this dataset.

High-risk profiles to prioritize (Use these as rule-based targeting in the short term to capture most at-risk customers) - Primary rule (high precision): Geography == Germany AND Age >= 40 AND IsActiveMember == 0 - Rationale: Germany has high churn base-rate and older inactive customers show the highest risk. - Alternative broader rule (higher recall): Age >= 40 AND Balance >= 80,000 AND IsActiveMember == 0 - Rationale: captures high-value, inactive older customers who are likely to churn.

Model interpretation & limitations - Feature importances indicate Age and economic indicators (salary, balance, credit score) matter most. Activity (IsActiveMember) is actionable (lower importance numerically but important in subgroup differences). - Model AUC is strong but recall on churn is low (0.45) — many actual churners are missed. This suggests class imbalance and/or limited features (no behavioral/transactional data, complaints, web/mobile usage, support interactions). - Recommendations to improve model: add transactional features, session/usage metrics, complaint logs; use class weighting, SMOTE or other resampling; tune threshold to increase recall (tradeoff precision).

Recommended actions (short, medium, long term) Short-term (weeks) — low cost, quick wins 1. Run targeted retention campaigns to high-risk segments - Germany-focused outreach: priority phone/email campaigns for German customers flagged high-risk. - Re-engagement for inactive customers: push notifications, emails offering personalized product suggestions, limited-time offers. - Incentivize with fee waivers, higher interest on savings for a limited period, or loyalty points for existing high-balance customers. 2. Proactively contact high-value accounts (Balance >= $80k, Age >= 40) with dedicated relationship managers or premium service line. 3. Analyze customer support logs for German customers and older cohorts for common complaints; triage and fix quick wins.

Medium-term (1–3 months) 1. Cross-sell / bundle strategy: customers with fewer products are somewhat more likely to churn — offer tailored bundles (e.g., savings + credit card + insurance) with incentives for adoption. 2. Improve onboarding and reactivation flows to increase IsActiveMember metric (gamification, clearer product benefits, nudges). 3. A/B test retention offers to identify cost-effective incentives; measure incremental retention and CLV uplift.

Long-term (3–12 months) 1. Enhance data collection: instrument app/website to capture usage patterns, incorporate transactional frequency, complaints, NPS, and marketing touchpoints. 2. Build a production propensity-to-churn model with explainability (SHAP) and uplift modeling for personalized offers — optimize for economic value (expected CLV delta vs retention cost). 3. Introduce loyalty tiers and relationship management for high-balance and older segments.

Implementation checklist & KPIs - Short-term KPI targets (6–12 weeks): reduce monthly churn among targeted Germany cohort by 15%; increase activity rate among targeted inactive customers by 25%. - Medium-term KPIs (3 months): increase cross-sell rate by 10% among targeted customers; improve model recall to 0.65 while maintaining precision >= 0.6. - Long-term KPIs (6–12 months): increase overall retention rate by 5 percentage points; increase average customer LTV of retained customers by 10%.

Operational playbook (example SQL/pandas filter to create target list) - High-precision target (short-term): df_target = df[(df['Geography']=='Germany') & (df['Age']>=40) &

(df['IsActiveMember']==0)] - Broader target (higher recall): df_target = df[(df['Age']>=40) & (df['Balance']>=80000) & (df['IsActiveMember']==0)]

Model improvement suggestions (technical) - Resampling: try SMOTE or class weights to lift recall for class=1. - Threshold tuning: move probability threshold lower to favor recall, then use business cost to choose threshold. - Feature engineering: create RFM-style features, monthly balances/changes, number of logins, product tenure, complaint counts. - Explainability: run SHAP values to understand per-customer drivers.

Limitations - Dataset lacks behavioral and support data which are often key churn drivers. - Causality not proven; associations are suggestive and should drive experiments (A/B tests) before large-scale rollouts.

Next analytical steps 1. Obtain transaction and engagement data and retrain model. 2. Perform cohort and survival analysis to understand timing of churn. 3. Implement SHAP-based explanations and build a retention-treatment policy using uplift modeling. 4. Set up dashboards to monitor churn by country, age bucket, balance buckets, and product mix.

Appendix (key numbers) - Rows: 10,000; churners: 2,037 (20.37%). - Average age: churners 44.84 vs non-churners 37.41. - Average balance: churners $91,108 vs non-churners $72,745. - Country churn rates: Germany ≈ 32.5%, France ≈ 16.2%, Spain ≈ 16.7%. - Model AUC: 0.851; churn recall: 0.452; churn precision: 0.760.

If you want, I can: - Generate targeted lists (CSV) using the rules above. - Re-run the model optimizing for recall and produce new metrics. - Produce visualizations (histograms, lift charts, SHAP plots) to include in an expanded PDF.